

7.3 Compression and Deployment

Prune, quantize, distill, then speculative decoding at serving time.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Scaling (note 7.1) makes a model that may not fit on the box you have. **Efficiency** is how you ship it: fewer weights, fatter arithmetic, a smaller student, or fewer expensive forward passes at decode time.

1. Shrink the weights

Prune: set small weights to zero (unstructured) or drop heads, neurons, or layers (structured). You need a pass of recovery training or the quality drop is ugly.

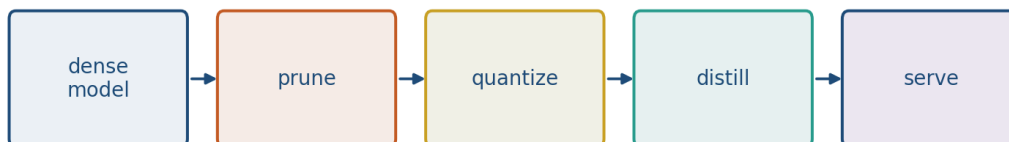
Quantize: store weights (and sometimes activations) in fewer bits. INT8 and 4-bit GPTQ/AWQ are the ones you will see in Hugging Face. Affine quantization uses a **scale** and **zero-point**:

$$x \approx s(q - z), \quad q \in \{0, \dots, 2^b - 1\}.$$

Lab 7 does this on a toy tensor and reports MSE. The curve is not free: outliers in activations break naive INT8.

Distill: a frozen **teacher** produces logits or hidden states; a smaller **student** matches them (plus the usual labels). You buy a model that is cheaper to run, not a new capability.

Compression is a pipeline. Measure quality after each cut.



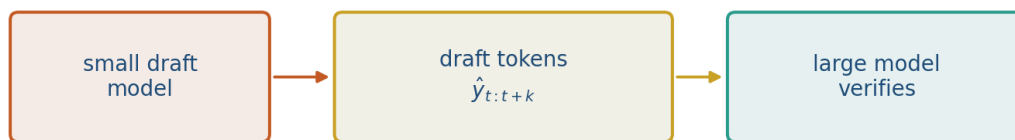
$s = 2/255$, $z = 128$. Drop z and the map cannot go below 0.

INT8 on $[-1, 1]$	value
$q = 128$	$s(q - 128) = 0$
$q = 255$	$s \cdot 127 \approx 0.996$
no z	cannot represent negatives

2. Speculative decoding

Autoregressive decode is serial: one token, then another. **Speculative decoding** runs a cheap **draft** model for several tokens, then the large **target** checks them in one parallel pass. If the target agrees, you accept a block; if not, you roll back at the first mismatch and sample from the target. Expected speedup is $> 1 \times$ when the draft is often right; the **distribution** is still the target's.

Speculative decoding: guess many tokens, accept the prefix the big model agrees on



This is not pruning. It is extra compute on a small net to save steps on a large net.

If the draft proposes 4 tokens and the target accepts 3, you paid one large forward for 3 tokens instead of 3 large forwards. Token 4 is resampled from the target.

3. What to report

Name bits, which layers you quantized, whether you distilled, and tokens per second versus a dense fp16 baseline. Quality: the same eval as the uncompressed model, not only perplexity. For a project, a 4-bit laptop demo is a deployment result; it is not a new architecture.

Stanford CS224N 2025 L11 puts **pruning** and **LoRA** in the same PEFT hour: fewer weights, or a small Δ on a frozen W . Distill and speculative decoding stay on this board; they are not LoRA. Note **8.3** owns

the BA algebra.

4. Teaching this note

About **35 minutes** at the board: affine INT8 on a 1-D range, a 4-token speculative accept/reject, then one sentence on prune vs distill. Play LoRA **0:00--~15:00** as “deploy a small delta,” then return to prune/quantize/speculate. Lab 7’s toy quantize is studio, not this block.

5. Worked example

Weights in $[-1, 1]$, $b = 8$ bits, $q \in \{0, \dots, 255\}$. A symmetric affine map:

$$s = \frac{2}{255}, \quad z = 128, \quad x \approx s(q - 128).$$

$q = 128 \mapsto 0$; $q = 255 \mapsto s \cdot 127 \approx 0.996$. If you drop z and only use $x \approx sq$, you cannot represent negatives.

Speculative: draft emits (t_1, t_2, t_3, t_4) . Target, in one forward, assigns probabilities. It agrees on t_1, t_2 , rejects t_3 . You commit t_1, t_2 , sample a new t'_3 from the **target**, and throw t_4 away. Speedup ≈ 2 large-tokens per large-forward this step, if the draft is that accurate on average.

Memory cartoon: 7B fp16 $\approx$ 14 GB. INT4 weights $\approx$ 3.5 GB plus overhead. Same architecture, different bits.

6. Where students get stuck

- Quantizing without a zero-point when the tensor is not ≥ 0 .
 - Thinking speculative decoding **changes** the target distribution. It must not; rejected drafts are resampled from the target.
 - Reporting perplexity only after 4-bit. The project eval is the same downstream number as fp16.
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7. Video

[Umar Jamil — LoRA explained](#). Play about **0:00–15:00** (frozen W , small BA). That is the deploy-small-deltas story. **Prune, quantize, and speculative decoding stay on the board**; they are not in this clip. Full LoRA is note **8.3**.

8. Practice

1. Why does a scale-and-zero-point map need z , not only s , if x is not centered at zero?
2. Distillation needs a teacher. What happens if the teacher is already calibrated badly on your domain?
3. Speculative decoding must not change the target distribution. Where does a rejected draft token get replaced?
4. Map $x \in [-2, 2]$ to INT8 with $q \in \{0, \dots, 255\}$. Give s and z so that $q = 0$ is -2 and $q = 255$ is 2 .
5. A draft proposes 5 tokens; the target accepts the first 4. How many **target** forwards did you pay for those 4 tokens, and what happens to token 5?